



Deadly nightshade
A perennial with large leaves and black berries, growing to 5 ft (1.5 m).

Atropa belladonna (Solanaceae)

Deadly Nightshade, Belladonna

Although deadly nightshade conjures up images of poison and death, like many plants it is an important and beneficial remedy when used correctly. Some of its constituents are employed in conventional medicine, for example to dilate the pupils for eye examinations and as an anesthetic. In herbal medicine, deadly nightshade is mainly prescribed to relieve intestinal colic and to treat peptic ulcers.



Deadly nightshade produces unmistakable cherry-sized, glossy black berries in autumn.

Habitat & Cultivation

Deadly nightshade is native to Europe, western Asia, and northern Africa, and is now cultivated worldwide. It thrives in chalky soils, in woods, and in open areas. The leaves are harvested in summer, and the root is collected from the first year onward in autumn.

Related Species

Many of the Solanaceae family are powerful medicines, including eggplant (*Solanum melongena*, p. 270), tobacco (*Nicotiana tabacum*, p. 240), and henbane (*Hyoscyamus niger*, p. 220).

Key Constituents

- Tropane alkaloids (up to 0.6%), including hyoscyamine and atropine
- Flavonoids
- Coumarins
- Volatile bases (nicotine)

Key Actions

- Smooth muscle antispasmodic
- Narcotic

- Reduces sweating
- Sedative

Research

■ **Tropane alkaloids**, The action of the tropane alkaloids is well understood. They inhibit the parasympathetic nervous system, which controls involuntary bodily activities. This reduces saliva; gastric, intestinal, and bronchial secretions; as well as the activity of the urinary tubules, bladder, and intestines. Tropane alkaloids also increase heart rate and dilate the pupils.

Traditional & Current Uses

■ **Folklore** Deadly nightshade was believed to help witches fly. Its other name “belladonna” (beautiful woman) is thought to refer to its use by Italian women to dilate the pupils of their eyes, making them more attractive.

■ **Relaxant** Deadly nightshade has been used in the same way throughout history. It is prescribed to relax distended organs, especially the stomach and intestines, relieving intestinal colic and pain. It helps peptic ulcers by reducing gastric acid production, and it relaxes spasms of the urinary tubules.

■ **Parkinson's disease** The herb can be used to treat the symptoms of Parkinson's disease, reducing tremors and rigidity, and improving speech and mobility.

■ **Anesthetic** muscle-relaxant properties of deadly nightshade make it useful in conventional medicine as an anesthetic, particularly when digestive or bronchial secretions need to be kept to a minimum.

Parts Used

Leaves are harvested in early summer. They have a weaker action than the root, and are more commonly used.



Dried leaves

Leaves, like the root, have relaxant properties.



Fresh leaves



Fresh root



Dried root

Root is collected in autumn.

Key Preparations & Their Uses

⚠ **Cautions** Take only if prescribed by a medical herbalist or doctor. Deadly nightshade can be fatal if taken at the wrong dosage.



Tincture, made from the leaves or the root, is a strong relaxant. It is prescribed by herbal practitioners to relieve colic and to treat Parkinson's disease.